BIJOY SARKAR

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Date of Birth	Nationality	Gender	LinkedIn	GitHub	Portfolio
19-10-2005	INDIAN	Male	bijoysarkar19	BijoySarkar19	bijoy-sarkar-portfolio

PROFILE

Computer Science & Engineering undergraduate with hands-on experience in AI/ML, full-stack web development, cybersecurity, and mobile application development. Experienced in Python, Flutter, Firebase, computer vision, NLP, and software development through internships and practical projects. Currently seeking software development and AI/ML opportunities to build production-oriented solutions and strengthen problem-solving skills.

EDUCATION

Bachelor of Technology in Computer Science and Engineering **Aug 2023 – Present**
ICFAI University Tripura, Agartala

WORK EXPERIENCE

Cybersecurity Intern **Jun 2026 – Jul 2026**
ETDC, STQC Directorate, MeitY, Government of India

- Worked on cybersecurity frameworks and Generative AI security with focus on security assessment of AI applications.
- Conducted security risk analysis using **OWASP guidelines for Large Language Models (LLMs)** to identify potential vulnerabilities and mitigation strategies.
- Gained practical exposure to vulnerability assessment, AI application security, and IT testing and quality assurance practices.

AI/ML Intern **Jan 2026 – Mar 2026**
National Institute of Technology Silchar

- Worked on a **Road Segmentation** project using satellite imagery for identifying and extracting road networks.
- Performed image preprocessing, dataset preparation, feature extraction, and semantic segmentation using Python.
- Worked with satellite imagery datasets and evaluated computer vision models for road detection and segmentation.

App Developer Intern **Jun 2025 – Jul 2025**
TalkEng, Agartala

- Developed an **Expense Tracker** mobile application using **Flutter and Dart**.
- Implemented Firebase Authentication and real-time database integration for secure user data management.
- Worked on application functionality, user interface development, and overall user experience.

Full-Stack Development Intern **Jun 2024 – Aug 2024**
Apni Vidya, Agartala

- Contributed to both front-end and back-end development of web applications.
- Worked on application design, coding, testing, debugging, and deployment-related activities.
- Collaborated on feature development and improved application functionality through practical development tasks.

Computer Tutor **Mar 2022 – Aug 2022**
Novel Computer, Agartala

- Taught computer fundamentals, programming basics, MS Office, and practical IT skills through hands-on learning.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, Dart, SQL

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, OpenCV, Jupyter Notebook

Web & App: HTML, CSS, Flask, Streamlit, Flutter, Firebase

Databases: MySQL, SQLite, Firestore

Cybersecurity: OWASP, LLM Security, Vulnerability Assessment, AI Security

Tools: Git, GitHub, Android Studio, VS Code

PROJECTS

OWASP AI Security Assistant **[GitHub]**
Python | Flask | Ollama | Llama 3 | Docker | OWASP

- Developed a cybersecurity web application to assess code, URL endpoints, and infrastructure against OWASP vulnerability categories.
- Integrated local AI inference using **Ollama and Llama 3** for security analysis, vulnerability assessment, and remediation recommendations.
- Implemented assessment history, security metrics, dataset management, re-evaluation workflows, and Docker Compose deployment.

Rakshak-AI **[GitHub]**
Python | LM Studio | Llama 3.2 | Tkinter | SQLite

- Developed a locally hosted AI assistant using **Python** and LM Studio.
- Integrated conversational assistance, desktop GUI, BMI calculation, speech recognition, and text-to-speech functionality.
- Used **SQLite** for persistent data management and an interactive assistant workflow.

Road Segmentation from Satellite Images **[GitHub]**
Python | Computer Vision | Jupyter Notebook | Kaggle

- Developed a computer-vision workflow for identifying and extracting road networks from satellite imagery.
- Performed image preprocessing, dataset preparation, model training, and evaluation.
- Explored semantic segmentation for geographic mapping and infrastructure planning.

Expense Tracker **[GitHub]**
Flutter | Dart | Firebase | Firestore

- Developed a mobile application for recording and categorizing daily expenses.
- Integrated **Firebase Authentication and Firestore** for secure user authentication and real-time data management.
- Implemented expense summaries and an intuitive interface for monitoring personal spending.

Smart Attendance System Using Face Recognition **[GitHub]**
Python | OpenCV | Face Recognition | Computer Vision

- Developed an automated attendance system using **face recognition and computer vision**.
- Implemented face detection, feature extraction, identity matching, and automated attendance recording.
- Designed organized attendance history with secure administrative access and device-token checks.

CivicLens Core **[GitHub]**
FastAPI | PostgreSQL | PostGIS | Redis | Docker | Next.js

- Built a full-stack public-infrastructure intelligence platform for mapping assets and reported civic issues.
- Designed API, database, geospatial, caching, background-worker, and object-storage components with Docker.
- Structured Part 1 to support future AI-based risk prediction, prioritization, hotspot detection, and maintenance recommendations.

MiniC Compiler Studio **[GitHub]**
C/C++ | Compiler Design | Python | GCC | Docker

- Built a web-based compiler environment for a MiniC language with a browser-based development interface.
- Integrated lexical analysis, parsing, compilation workflows, GCC execution, and source-code processing.
- Containerized the application and exposed the development studio through a Dockerized web server.

PixelDB Engine **[GitHub]**
Java | Maven | Database Systems | SQL

- Developed a database engine from scratch to explore storage, query-processing, and database-system internals.
- Used Java and Maven with automated tests and a structured engine architecture.
- Planned a pixel-art themed interface to make database concepts more interactive and demonstrable.

Advanced NLP Analytics **[GitHub]**
Python | Flask | NLP | TF-IDF | Logistic Regression | Docker

- Built an end-to-end NLP application for text classification using preprocessing, tokenization, unigram/bigram TF-IDF features, and Logistic Regression.
- Added a Flask interface for real-time text analysis and CSV profiling with confidence scores and keyword extraction.
- Added automated tests, Docker support, and GitHub Actions-oriented project structure for reproducible development.

AI-Based Text Sentiment Analysis – SentimentX **[GitHub]**
Python | Flask | NLTK | VADER | YouTube API

- Developed a web-based sentiment analysis system using **VADER** and NLTK for positive, negative, and neutral classification.
- Integrated YouTube comment analysis through the YouTube API and automated scanning of comment samples.
- Built the application with Flask and structured it for containerization and GitHub-based deployment workflows.

NovaMart – Multi-Vendor Commerce Platform **[GitHub]**
Next.js | TypeScript | Prisma | SQLite | Razorpay

- Designed a full-stack multi-vendor commerce platform with product, checkout, seller, and customer workflows.
- Used Next.js, TypeScript, Prisma, and SQLite for application and database development.
- Integrated payment-oriented architecture and developed frontend interactions for the commerce experience.

Vaultora Bank – Digital Banking Platform **[GitHub]**
Digital Banking | UPI | Card Services | E-Rupee | Web Security

- Developed a digital-banking website concept covering core online banking and customer account workflows.
- Designed flows for debit and credit card services, UPI payments, and digital-rupee functionality.
- Included validation-focused interactions, privacy information, and a custom error experience as part of the website structure.

Aether – AI Engineering Platform **[GitHub]**
AI Engineering | Platform Development | Software Architecture

- Developed an AI engineering platform project focused on bringing AI-related development workflows into a structured application.
- Organized the project around reusable platform components and an extensible engineering approach.
- Documented the implementation in a dedicated repository for continued development and technical iteration.

Nexora – Private Search Engine **[GitHub]**
Private Search | AI-Assisted Search | Privacy-Focused Browsing

- Developed a privacy-oriented search application designed as an alternative to conventional search workflows.
- Explored AI-assisted search and browser-style discovery features with an emphasis on user control and private browsing.
- Structured search and information-discovery capabilities within a custom application experience.

JavaCacheDB **[GitHub]**
Java | Database Systems | Caching | Data Management

- Developed a Java-based database and caching project exploring data storage, retrieval, and cache-oriented access patterns.
- Organized the codebase around reusable components for managing stored data and cache operations.
- Used the project to explore database concepts and Java systems development.

LANGUAGES

English | Hindi | Bengali